

Topic 8

8A — Redox Chemistry

8.1 — What is an Oxidation Number?

Definition

The oxidation number (or oxidation state) of an atom is the **charge it would have** if the bonding in the compound were **completely ionic**.

8.2 — Rules for Assigning Oxidation Numbers

These rules must be applied in this **exact order** when determining unknown oxidation states:

Rule No.	Rule	Example
1	All elements in their uncombined (elemental) state have oxidation number 0	Na, Cl ₂ , O ₂ , Fe → 0
2	For simple ions , oxidation number = ionic charge	Na ⁺ → +1, O ²⁻ → -2
3	The sum of oxidation numbers in a neutral compound = 0	H ₂ O: 2(+1) + (-2) = 0
4	The sum of oxidation numbers in a polyatomic ion = charge on the ion	SO ₄ ²⁻ : S + 4(-2) = -2 ⇒ S = +6
5	Group 1 metals = +1, Group 2 metals = +2 in all compounds	Na in NaCl = +1, Mg in MgSO ₄ = +2

Note:

- You will often be asked to deduce unknown oxidation numbers using these steps backward.
- Apply rules in order** — e.g., oxygen takes -2 unless rule 1 or 2 overrides it.

8.3 — More Advanced Oxidation Number Rules

Rule No.	Rule	Example
6	Fluorine always has oxidation number -1 in compounds	F ₂ = 0, in HF → F = -1
7	Hydrogen is usually +1, but -1 in metal hydrides	H in HCl = +1, H in NaH = -1
8	Oxygen is usually -2, but:	
	• In peroxides (O ₂ ²⁻), each O = -1	H ₂ O ₂ : 2(-1) + 2(+1) = 0

	• In F_2O , O = +2 (F more electronegative)	
9	In combined halogens , the more electronegative halogen takes -1	$\text{ClF}_3 \rightarrow \text{F} = -1$, so $\text{Cl} = +3$
10	Oxidation numbers can be fractions in average state ions (e.g. disproportionation)	Fe_3O_4 : Fe = +2 and +3 average = $+8/3$

8.4 — Writing Formulae from Oxidation Numbers

You must be able to:

- **Construct a compound's formula** given oxidation states
- **Deduce unknown oxidation states** from a compound

Steps:

1. Identify oxidation numbers
2. Criss-cross the numbers (swap charges to make formula neutral)
3. Reduce to the **simplest ratio**

Examples:

Name	Step	Formula
Iron(III) oxide	Fe^{3+} and O^{2-}	Fe_2O_3
Chromium(VI) oxide	Cr^{6+} and O^{2-}	CrO_3
Tin(IV) chloride	Sn^{4+} and Cl^-	SnCl_4

8.5 — Oxidation and Reduction: Definitions and Dual View

Definition Type	Oxidation	Reduction
Electron transfer	Loss of electrons (OIL)	Gain of electrons (RIG)
Oxidation number	Increase in oxidation number	Decrease in oxidation number

Mnemonic: OIL RIG

Oxidation Is Loss (of electrons), Reduction Is Gain (of electrons)

Example (Fe^{2+} and MnO_4^-):

- $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^- \rightarrow$ Oxidation (loss of e^- , ox. no. increases $+2 \rightarrow +3$)
- $\text{MnO}_4^- + 5\text{e}^- \rightarrow \text{Mn}^{2+} \rightarrow$ Reduction (gain of e^- , ox. no. decreases $+7 \rightarrow +2$)

8.6 — Oxidising and Reducing Agents

Definitions:

- **Oxidising agent:** A substance that **oxidises** another by **gaining electrons** itself
- **Reducing agent:** A substance that **reduces** another by **losing electrons** itself

In other words:

Agent Type	Electron Change	Effect on Other Substance	Is It Itself Oxidised or Reduced?
Oxidising agent	Gains electrons	Causes oxidation	Itself reduced
Reducing agent	Loses electrons	Causes reduction	Itself oxidised

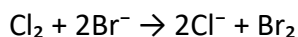
Examples:

- **Oxidising agents:** Acidified KMnO_4 , acidified $\text{K}_2\text{Cr}_2\text{O}_7$, Cl_2 , H_2O_2
- **Reducing agents:** Zn , Fe^{2+} , SO_2 , CO , NaBH_4

You must state whether a species acts as an oxidising or reducing agent based on:

- Electron transfer
- Oxidation number change

Example:



Chlorine is the oxidising agent; it causes Br^- to be oxidised and is itself reduced.

8.7 — Disproportionation

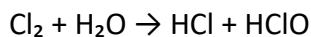
Definition:

A **disproportionation reaction** is one in which the **same element is both oxidised and reduced**, forming two products with different oxidation numbers.

Key signs:

- One reactant
- Same element appears in two products in different oxidation states

Common example:



Cl_2 is both:

- Reduced: $\text{Cl}_2 \rightarrow \text{HCl}$ ($0 \rightarrow -1$)
- Oxidised: $\text{Cl}_2 \rightarrow \text{HClO}$ ($0 \rightarrow +1$)

Other examples:

- $\text{Cl}_2 + \text{cold NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$
- $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$

8.8 — Using Oxidation Numbers to Classify Reactions

You may be asked to:

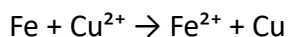
- Identify whether a reaction is redox
- Determine what is oxidised/reduced
- State the oxidising and reducing agents

Method:

1. Assign oxidation numbers to all elements
2. Look for:
 - Increase in oxidation number $\rightarrow$ oxidation
 - Decrease in oxidation number $\rightarrow$ reduction

If both occur, the reaction is **redox**

Example:



Fe: $0 \rightarrow +2$ (oxidised)

Cu^{2+} : $+2 \rightarrow 0$ (reduced)

This is a redox reaction.

8.9 — General Redox Trends of Metals and Non-Metals

Type	Electron Movement	Common Reaction	Charge of Ions Formed	Oxidation Number Trend
Metals	Lose electrons	Oxidised	Positive ions	Ox. number increases
Non-metals	Gain electrons	Reduced	Negative ions	Ox. number decreases

Examples:

- $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$ (oxidation)
- $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ (reduction)

Metals act as **reducing agents**

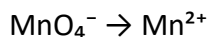
Non-metals act as **oxidising agents**

8.10 — Writing Half Equations and Full Ionic Equations

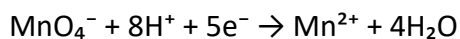
Steps to write half-equations (especially in aqueous/acidic solutions):

1. **Balance atoms** (except H and O)
2. **Balance O** using H_2O
3. **Balance H** using H^+
4. **Balance charge** using e^-

Example 1: MnO_4^- to Mn^{2+}



Balance:

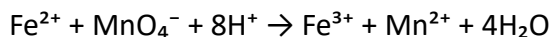


Example 2: Fe^{2+} to Fe^{3+}



To get the full redox equation:

- Multiply half-equations to balance e^-
- Add the two half-equations
- Cancel electrons and anything else duplicated on both sides



8B — The Elements of Groups 1 and 2

8.11 — Trend in Ionisation Energy Down Group 1 and Group 2

Ionisation energy is the enthalpy change when **1 mole of electrons** is removed from **1 mole of gaseous atoms** to form 1 mole of gaseous positive ions.

Example:



Trend Down the Group (applies to both Group 1 and Group 2):

Factor	Trend Down the Group	Explanation
Atomic radius	Increases	More electron shells
Shielding	Increases	Inner electrons block nuclear attraction
Nuclear attraction for outer electron	Decreases	Distance and shielding outweigh increase in nuclear charge
Ionisation energy	Decreases	Less energy required to remove electrons
Ease of oxidation	Increases	Electrons lost more easily
Reactivity	Increases	Faster and more vigorous reactions

As you go down either Group 1 or Group 2 in the periodic table, each successive element has an extra **electron shell**, which leads to an **increase in atomic radius**. Although the **nuclear charge** (number of protons) also increases, the additional inner shells cause **greater electron shielding**, which reduces the net attractive force between the nucleus and the outermost electron. Because the outer electron is both **farther away** and more effectively shielded, it is held **less strongly**. As a result, **less energy is required** to remove it. Therefore, the **first ionisation energy decreases** progressively down both groups, even though the nucleus becomes more positively charged.

Group Comparison:

Group 1 (e.g., Na, K)	Group 2 (e.g., Mg, Ca)
Only 1 electron to remove	2 electrons to remove (successively)
Lower first I.E. compared to G2	Higher first I.E. due to increased nuclear attraction

8.12 — Trends in Reactivity Down Group 1 and Group 2

Definition (Redox-based):

Reactivity in metals refers to how readily an atom **loses electrons** to form positive ions. The easier it is to lose outer electrons, the more reactive the element is.

In redox terms:

- **Group 1 and Group 2 metals are reducing agents**
- Reactivity = tendency to undergo **oxidation**: $M \rightarrow M^+ + e^-$ (or $2e^-$ for Group 2)

General Trend:

- Reactivity increases down both groups

Group-Specific Differences:

Group 1:

- React with water to form **alkaline solutions** and **hydrogen gas**
- Reactions become **increasingly vigorous** down the group

Example reactions:

- $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
- K reacts faster and more violently than Na

Group 2:

- React with water **less vigorously** than Group 1
- Reactivity still increases down the group, especially noticeable in reactions with **water** and **acids**

Example:

- Mg reacts slowly with cold water but rapidly with steam:
 $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$
- Ca, Sr, Ba react more readily even with cold water:
 $\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$

“Explain why Group 2 metals become more reactive down the group.”

Markscheme keywords:

1. Atomic radius increases and shielding increases
2. Nuclear attraction to outer electrons decreases
3. Outer electrons are lost more easily / lower ionisation energy

8.13 — Reactions of the Elements of Group 1 (Li to K) and Group 2 (Mg to Ba) with Oxygen, Chlorine, and Water

A. Reactions with Oxygen (O₂)

Group 1: Form simple metal oxides

Metal	Reaction Equation	Product	Notes
Li	$4\text{Li} + \text{O}_2 \rightarrow 2\text{Li}_2\text{O}$	Lithium oxide	White solid
Na	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$	Sodium oxide	Burns with yellow flame
K	$4\text{K} + \text{O}_2 \rightarrow 2\text{K}_2\text{O}$ or K_2O_2	Potassium oxide or peroxide	Lilac flame, vigorous

Note: As you go down Group 1, formation of **peroxides and superoxides** becomes more likely due to larger ionic radius and better stabilization of larger anions (O₂²⁻, O₂⁻).

Group 2: Form basic oxides

Metal	Reaction Equation	Product	Notes
Mg	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	Magnesium oxide	Burns with bright white flame
Ca	$2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$	Calcium oxide	Reacts more vigorously than Mg
Ba	$2\text{Ba} + \text{O}_2 \rightarrow 2\text{BaO}$	Barium oxide	Highly reactive with air

B. Reactions with Chlorine (Cl₂)

Group 1 and 2 metals both form white ionic chlorides

Group	Example Reaction	Product
Group 1	$2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$	Sodium chloride (NaCl)
Group 2	$\text{Mg} + \text{Cl}_2 \rightarrow \text{MgCl}_2$	Magnesium chloride (MgCl ₂)

- All halides formed are **soluble in water** (except BeCl₂, which is covalent)
- Reactivity increases down the group → reaction becomes more vigorous

C. Reactions with Water (H₂O)

Group 1:

Metal	Reaction	Product	Observation
Li	$2\text{Li} + 2\text{H}_2\text{O} \rightarrow 2\text{LiOH} + \text{H}_2$	Lithium hydroxide + H ₂	Fizzing, floats
Na	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$	Sodium hydroxide + H ₂	Melts into ball, yellow flame
K	$2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$	Potassium hydroxide + H ₂	Lilac flame, sparks, violent

- **Alkaline solutions** are formed (Group 1 hydroxides are strongly basic)
- Reaction becomes **more vigorous down the group**

Group 2:

- Reactivity with water **increases down the group**
- Products: **metal hydroxide + hydrogen gas**

Metal	Reaction	Observation
Mg	$\text{Mg} + \text{H}_2\text{O} (\text{cold}) \rightarrow$ very slow/no reaction	Reacts faster with steam: $\text{Mg} + \text{H}_2\text{O}(\text{g}) \rightarrow \text{MgO} + \text{H}_2$
Ca	$\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{H}_2$	Effervescence, white precipitate (sparingly soluble Ca(OH) ₂)
Ba	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	Vigorous fizzing, clear solution (Ba(OH) ₂ is soluble)

Group	Reactivity with Water	Nature of Hydroxide Formed
1	Very reactive	Soluble, strongly alkaline
2	Less reactive (increases down group)	Solubility increases down group

8.14 — Reactions of Group 1 and 2 Oxides and Hydroxides

8.14(i) — Reactions of Oxides with Water and Dilute Acids

A. Reaction of Metal Oxides with Water

Group 1 and Group 2 metal oxides are basic oxides.

They react with water to form **alkaline solutions** of **metal hydroxides**.

Oxide	Reaction with Water	Product Formed	Notes
Na ₂ O (Group 1)	Na ₂ O + H ₂ O → 2NaOH	Sodium hydroxide (soluble)	Strongly basic
MgO (Group 2)	MgO + H ₂ O → Mg(OH) ₂	Magnesium hydroxide (sparingly soluble)	Weak alkali
CaO (Group 2)	CaO + H ₂ O → Ca(OH) ₂	Calcium hydroxide (slightly soluble)	Limewater

Trend in Group 2:

- Solubility of hydroxides **increases down the group**
- Therefore, the **alkalinity of the solution increases** down the group

B. Reaction of Metal Oxides with Dilute Acids

These are classic **acid-base reactions**:

Metal oxide (base) + acid → salt + water

Oxide	Reaction with HCl	Product
Na ₂ O	Na ₂ O + 2HCl → 2NaCl + H ₂ O	Salt + water
MgO	MgO + 2HCl → MgCl ₂ + H ₂ O	Magnesium chloride + water
CaO	CaO + 2HNO ₃ → Ca(NO ₃) ₂ + H ₂ O	Calcium nitrate + water

These reactions **neutralise the acid**, showing the **basic nature** of metal oxides.

8.14(ii) — Reactions of Hydroxides with Dilute Acids

Hydroxides behave like bases and undergo **neutralisation reactions**:

Metal hydroxide + acid → salt + water

Hydroxide	Reaction with HCl	Product
NaOH	NaOH + HCl → NaCl + H ₂ O	Salt + Water
Mg(OH) ₂	Mg(OH) ₂ + 2HCl → MgCl ₂ + 2H ₂ O	Salt + water
Ba(OH) ₂	Ba(OH) ₂ + H ₂ SO ₄ → BaSO ₄ + 2H ₂ O	Precipitate of BaSO ₄ formed

Group 2 hydroxides:

- React similarly to Group 1 hydroxides
- Less soluble** → some hydroxides (e.g., Mg(OH)₂) may form **suspensions** instead of clear solutions
- Ba(OH)₂ is **highly soluble**, gives clear alkaline solution

Exam Notes:

- Always specify **products** in the correct ionic state
- Use **state symbols** when asked:
e.g., $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)}$
- For precipitates: note **insolubility** (e.g., BaSO_4)

8.15 — Trends in Solubility of Group 2 Hydroxides and Sulfates

A. General Patterns

Compound Type	Solubility Trend Down Group 2	Explanation
Hydroxides (M(OH)_2)	Increases down the group	Lattice enthalpy decreases more than hydration enthalpy
Sulfates (MSO_4)	Decreases down the group	Hydration enthalpy decreases faster than lattice enthalpy

B. Group 2 Hydroxides: M(OH)_2

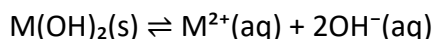
Solubility Trend:

- $\text{Mg(OH)}_2 \rightarrow$ very slightly soluble
- $\text{Ca(OH)}_2 \rightarrow$ sparingly soluble (forms limewater)
- $\text{Sr(OH)}_2 \rightarrow$ soluble
- $\text{Ba(OH)}_2 \rightarrow$ very soluble (strong alkaline solution)

Explanation:

- As ionic radius increases, the **lattice enthalpy** becomes less negative (weaker attraction between M^{2+} and OH^-)
- Hydration enthalpy also decreases, but **not as fast**
- So, **ΔH solution becomes more negative** $\rightarrow$ increased solubility

Equation:



Observation:

- Ba(OH)_2 gives a clear alkaline solution
- Mg(OH)_2 forms a white suspension

C. Group 2 Sulfates: MSO_4

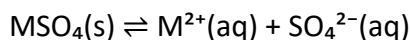
Solubility Trend:

- $\text{MgSO}_4 \rightarrow$ very soluble
- $\text{CaSO}_4 \rightarrow$ sparingly soluble
- $\text{SrSO}_4 \rightarrow$ poorly soluble
- $\text{BaSO}_4 \rightarrow$ virtually insoluble

Explanation:

- Larger cations form **weaker hydration enthalpies**
- Lattice energy does not decrease as much $\rightarrow \Delta H$ solution becomes less negative or even positive
- So, solubility **decreases** down the group

Equation:



D. Practical Use: BaSO_4 in Medical Imaging

- **Barium sulfate is insoluble**, so it is used as a **barium meal** in X-ray imaging
- **Safe** despite toxicity of Ba^{2+} because it does **not dissolve**, so Ba^{2+} is not released into the body

Summary Table

Compound	Mg^{2+}	Ca^{2+}	Sr^{2+}	Ba^{2+}
$\text{M}(\text{OH})_2$ Solubility	Low	Moderate	High	Very High
MSO_4 Solubility	High	Moderate	Low	Very Low (Insoluble)

8.16 — Thermal Stability of Group 1 and 2 Nitrates and Carbonates

A. What is Thermal Stability?

Definition

Thermal stability refers to a compound's **resistance to decomposition upon heating**.

A thermally **stable** compound:

- Requires **more heat** to decompose

- Decomposes at a **higher temperature**

B. Reactions Upon Heating

1. Carbonates (MCO_3)

- **Group 1** (except Li): do **not** decompose easily
- **Group 2**: decompose to form metal oxide and carbon dioxide

General reaction (Group 2):



Example:



2. Nitrates (MNO_3)

- **Group 1**: decompose to **nitrite + O_2** (except Li)
- **Group 2**: decompose to **metal oxide + NO_2 + O_2**

Group	General Reaction
Group 1	$2\text{MNO}_3 \rightarrow 2\text{MNO}_2 + \text{O}_2$ (Li is an exception)
Group 2	$2\text{M}(\text{NO}_3)_2 \rightarrow 2\text{MO} + 4\text{NO}_2 + \text{O}_2$

Examples:

- $\text{NaNO}_3 \rightarrow \text{NaNO}_2 + \frac{1}{2}\text{O}_2$
- $\text{Mg}(\text{NO}_3)_2 \rightarrow \text{MgO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$

Observation:

- Brown gas (NO_2) released during Group 2 nitrate decomposition

C. Thermal Stability Trend

Group 2:

- Thermal stability of **carbonates and nitrates increases down the group**
 - $\text{MgCO}_3 < \text{CaCO}_3 < \text{SrCO}_3 < \text{BaCO}_3$
 - $\text{Mg}(\text{NO}_3)_2 < \text{Ca}(\text{NO}_3)_2 < \text{Sr}(\text{NO}_3)_2 < \text{Ba}(\text{NO}_3)_2$

D. Explanation of the Trend (Cation Size and Charge)

Factor	Explanation
Cation size increases	Down the group
Charge stays the same	M ²⁺ for Group 2
Charge density decreases	Weaker polarising power
Polarisation of anion decreases	Carbonate/nitrate anion less distorted
More stable compound	Harder to decompose

Summary:

- Small, highly charged cations (e.g. Mg²⁺) have **high polarising power**
- They distort the electron cloud of CO₃²⁻ or NO₃⁻ → **less stable**
- Larger cations (e.g. Ba²⁺) have less effect → **more stable**

E. Group 1 Exceptions

- Li⁺ is small with high charge density (like Group 2)
- So, Li₂CO₃ and LiNO₃ decompose like Group 2 compounds

Examples:

- Li₂CO₃ → Li₂O + CO₂
- 4LiNO₃ → 2Li₂O + 4NO₂ + O₂

8.17 — Flame Colours in Group 1 and Group 2 Compounds

A. What Causes Flame Colours?

Explanation:

When a compound is heated in a flame:

1. **Electrons in metal ions gain energy** and are promoted to **higher energy levels** (excited state).
2. These electrons are **unstable** and quickly fall back down to their **original (ground) state**.
3. As they return, they **release energy as visible light**.
4. The **wavelength (colour)** of light corresponds to the **energy gap** between levels.

B. Flame Colours of Group 1 Elements

Element	Flame Colour
Li	red
Na	Bright yellow
K	Lilac (pale violet)
Rb	Red-violet
Cs	Blue-violet

- **Na's yellow** flame is very intense and can **mask** other colours
- Use a **cobalt blue glass** to filter out Na yellow in tests

C. Flame Colours of Group 2 Elements

Element	Flame Colour
Ca	Brick red
Sr	Crimson red
Ba	Apple green
Mg	No colour

- **Mg²⁺** gives no flame colour because its excitation energy is **too high** for visible light
- Flame colours come from the **metal ion**, not the anion

D. How to Perform a Flame Test

Steps:

1. Dip a clean platinum or nichrome wire in concentrated HCl
2. Hold it in a roaring Bunsen flame until no colour shows (clean)
3. Dip into powdered sample
4. Place in edge of flame and observe colour

Precautions:

- Clean the wire between each sample
- Use **concentrated HCl** to form volatile metal chlorides (enhances flame colour)

8.18(i) — Investigating Thermal Decomposition of Group 1 and Group 2 Nitrates and Carbonates

A. Purpose of the Experiment

To observe how **thermal stability varies** across Group 1 and Group 2 compounds by heating their **nitrates or carbonates** and identifying decomposition products.

B. Decomposition Reactions

Carbonates:

- Group 2:
 $\text{MCO}_3(\text{s}) \rightarrow \text{MO}(\text{s}) + \text{CO}_2(\text{g})$
(except Group 1 which don't decompose, except Li_2CO_3)

Nitrates:

- Group 2:
 $2\text{M}(\text{NO}_3)_2 \rightarrow 2\text{MO} + 4\text{NO}_2 + \text{O}_2$
- Group 1 (except Li):
 $2\text{MNO}_3 \rightarrow 2\text{MNO}_2 + \text{O}_2$

C. Procedure for Thermal Decomposition

- Place a **small amount of solid** in a hard glass test tube
- Insert a delivery tube into **limewater** or hold a glowing splint nearby
- Heat the solid strongly
- Observe:
 - Gas evolved (CO_2 or NO_2)
 - Condensation (for water)
 - Flame colours (for identification)

D. Expected Observations

Compound	Observation	Gas Detected
MgCO_3	White solid $\rightarrow$ white solid + CO_2	CO_2 turns limewater milky
CaCO_3	Same as above, requires stronger heating	CO_2
BaCO_3	May show little to no decomposition	Little or no gas
$\text{Mg}(\text{NO}_3)_2$	Brown gas (NO_2) and white solid residue	$\text{NO}_2 + \text{O}_2$
LiNO_3	Brown gas + white solid (Li_2O)	$\text{NO}_2 + \text{O}_2$

8.18(ii) — Identifying Gases Evolved in Decomposition

A. Carbon Dioxide (CO₂)

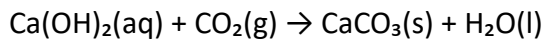
Test:

- Bubble through **limewater (Ca(OH)₂)**

Positive Result:

- Limewater turns **milky/cloudy** due to formation of CaCO₃

Equation:



B. Oxygen (O₂)

Test:

- Insert a **glowing splint** into the test tube

Positive Result:

- Splint **relights** in oxygen

C. Nitrogen Dioxide (NO₂)

Observation only (no test):

- NO₂ is a **brown gas**
- Has a **sharp, choking smell**
- Acidic gas — turns **moist universal indicator red**

D. Nitrogen Monoxide (NO) – Not normally tested

- Colourless
- Reacts quickly with oxygen in air to form brown NO₂

Safety Note

- NO₂ is **toxic and acidic**
- Use **fume hood** or well-ventilated area for heating nitrates

8.19 — Identification of Inorganic Ions: Group 1 and 2 Compounds

A. 8.19(i) — Test for Carbonate (CO_3^{2-}) and Hydrogencarbonate (HCO_3^-) Ions

Principle:

Both carbonate and hydrogencarbonate ions **release CO_2** when reacted with an **acid**.

Test Procedure:

1. Add **dilute hydrochloric acid** (or any dilute strong acid) to the solid or solution
2. Collect gas or bubble it through **limewater**

Positive Result:

- **Effervescence** (CO_2 gas)
- **Limewater turns milky**

Equations:

- $\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- $\text{HCO}_3^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- $\text{CO}_2(\text{g}) + \text{Ca}(\text{OH})_2(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$

Distinguishing Between CO_3^{2-} and HCO_3^- :

- Add **magnesium chloride solution** to both:
 - $\text{CO}_3^{2-} \rightarrow$ white precipitate (MgCO_3)
 - $\text{HCO}_3^- \rightarrow$ no precipitate unless solution is made alkaline first

B. 8.19(ii) — Test for Ammonium Ion (NH_4^+)

Principle:

NH_4^+ reacts with OH^- to release **ammonia gas (NH_3)** when warmed.

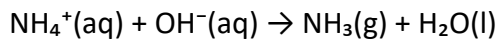
Test Procedure:

1. Add **sodium hydroxide solution** to the sample
2. Warm the mixture gently
3. Test gas released with:
 - **Damp red litmus paper** (turns blue)
 - **HCl fumes** (white smoke of NH_4Cl)

Positive Results:

- **Sharp smell of ammonia**
- **Red litmus turns blue**
- **White fumes** with concentrated HCl gas

Equation:



C. 8.19(iii) — Test for Sulfate Ion (SO_4^{2-})

Principle:

Sulfate ions form a **white precipitate** with **barium ions**, but only in **acidified conditions**.

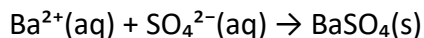
Test Procedure:

1. Add **dilute HCl** (to remove CO_3^{2-} interference)
2. Add **barium chloride solution** ($\text{BaCl}_2(\text{aq})$)

Positive Result:

- **White precipitate of barium sulfate** (BaSO_4)

Equation:



Important Notes:

- Acidification is essential to **prevent false positives** from carbonates or sulfites
- Do not use H_2SO_4 to acidify (would add sulfate ions)

8.20 — Calculating Solution Concentrations in mol dm^{-3} and g dm^{-3} , Including Titrations

A. Definitions

- **Concentration (mol dm^{-3}):** Number of moles of solute per dm^3 of solution

Formula:

$$c = n / V$$

where:

c = concentration (mol dm⁻³)

n = number of moles (mol)

V = volume (dm³)

- **Concentration (g dm⁻³):** Mass of solute per dm³ of solution

Formula:

$$c = m / V$$

where:

c = concentration (g dm⁻³)

m = mass of solute (g)

V = volume (dm³)

B. Converting Between Units

To convert **mol dm⁻³ to g dm⁻³:**

$$c \text{ (g dm}^{-3}\text{)} = c \text{ (mol dm}^{-3}\text{)} \times M_r$$

Example:

A 1.50 mol dm⁻³ solution of NaOH ($M_r = 40.0$)

$$\rightarrow \text{g dm}^{-3} = 1.50 \times 40.0 = \mathbf{60.0 \text{ g dm}^{-3}}$$

C. Common Molarity and Volume Conversions

From	To	Multiply/Divide by
cm ³	dm ³	÷ 1000
dm ³	cm ³	× 1000

Example:

$$25.0 \text{ cm}^3 = 0.0250 \text{ dm}^3$$

D. Titration Calculations

Used to determine unknown concentrations by reacting a solution of known concentration with one of unknown concentration until neutralisation.

1. General Titration Equation

acid + base → salt + water

From the balanced chemical equation:

- Determine mole ratio

- Use moles = concentration $\times$ volume
- Rearrange to find unknown

2. Typical Steps:

Given:

- Volume and concentration of one solution
- Volume of the other solution
- Find unknown concentration

Example:

25.0 cm³ of HCl neutralised by 20.0 cm³ of 0.100 mol dm⁻³ NaOH

Reaction: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

1. Moles of NaOH = $0.100 \times 0.0200 = 0.00200$ mol
2. Mole ratio HCl : NaOH = 1 : 1
 $\Rightarrow$ Moles of HCl = 0.00200 mol
3. Volume of HCl = 25.0 cm³ = 0.0250 dm³
4. $c = n / V = 0.00200 / 0.0250 = \mathbf{0.0800 \text{ mol dm}^{-3}}$

E. Indicators for Titrations

Indicator	Suitable Titration Type	Colour Change
Phenolphthalein	Strong base vs strong/weak acid	Pink $\rightarrow$ colourless
Methyl orange	Strong acid vs weak/strong base	Red $\rightarrow$ yellow

Note:

- Do not use universal indicator for titrations — colour change is too gradual

(8.21 and 8.23 is skipped as they are core practicals)

8.22 — Minimising Sources of Measurement Uncertainty in Volumetric Analysis

A. What is Uncertainty?

Uncertainty is the **possible error range** in a measurement.

It is caused by the limitations of measuring instruments or human error.

B. Instrumental Uncertainties (Per Reading)

Apparatus	Typical Uncertainty per Reading	Notes
Burette	$\pm 0.05 \text{ cm}^3$	But you use it twice per titration, so $\pm 0.10 \text{ cm}^3$ per titre
Pipette (25.0 cm^3)	$\pm 0.06 \text{ cm}^3$	Volumetric pipette
Measuring cylinder	$\pm 0.5\text{--}1.0 \text{ cm}^3$	Not suitable for titrations
Digital balance	$\pm 0.01 \text{ g}$ or $\pm 0.001 \text{ g}$	Based on model

C. How to Minimise Uncertainty

Method	How it Helps
Use more accurate equipment	Smaller uncertainty per reading
Take larger volume titres	Reduces % uncertainty because relative error decreases
Repeat titrations and take concordant results	Identifies outliers and improves reliability
Use equipment with finer scales	e.g., burette over measuring cylinder
Use a white tile	Better visibility of colour change
Add solution dropwise near end point	Avoid overshooting
Rinse apparatus properly	Prevent dilution or contamination

D. Calculating Percentage Uncertainty

Formula:

$$\% \text{ uncertainty} = (\text{absolute uncertainty} \div \text{measured value}) \times 100$$

Example 1 (burette):

Measured titre = 25.00 cm^3

Uncertainty per titre = $\pm 0.10 \text{ cm}^3$

$$\rightarrow \% \text{ uncertainty} = (0.10 \div 25.00) \times 100 = \mathbf{\pm 0.40\%}$$

Example 2 (weighing):

Mass of solid = 0.20 g

Balance uncertainty = $\pm 0.01 \text{ g}$

$$\rightarrow \% \text{ uncertainty} = (0.01 \div 0.20) \times 100 = \mathbf{\pm 5.0\%}$$

Important Tip:

- **Smaller values increase % uncertainty**
- So, **use more concentrated/voluminous solutions** when possible
- When a **measurement is taken 2 or more times**, the uncertainty is **multiplied** by the **number of times it is taken**. For example: **If the average measured titre is 24.00cm³ and the measurements are taken two times and the uncertainty per titre is $\pm 0.10 \text{ cm}^3$, then, %U = $(0.1 \times 2)/24.00 \times 100 = 0.83\%$**
- The earlier case arises due to the calculation rules for uncertainty, if the values are added or subtracted, the uncertainties are added, **here, the uncertainty for both value is the same** as the same apparatus is **used for measurement**.

8C — Group 7 Chemistry

8.24 — Group 7: Trends in Physical Properties and Reactivity

A. Physical State and Colour of Halogens at Room Temperature

Group 7 elements exist as **diatomic molecules (X₂)**.

Their physical state and colour change predictably down the group.

Element	State at Room Temp	Colour	In Solution (organic solvent)	Notes
Cl ₂	Pale green gas	Pale green	Pale green in cyclohexane	Poisonous gas
Br ₂	Brown liquid	Red-brown (vapour too)	Orange in cyclohexane	Volatile liquid
I ₂	Shiny grey-black solid	Purple vapour	Violet in cyclohexane	Sublimes when heated

B. Trends in Melting and Boiling Points

Melting and boiling points increase down the group.

Reason	Explanation
More electrons per molecule	Increases strength of London forces
Larger surface area	More area for instantaneous dipoles to form
Stronger intermolecular forces	Requires more energy to separate molecules

As a result:

- **Cl₂**: gas at room temp
- **Br₂**: liquid
- **I₂**: solid

C. Trends in Electronegativity

Electronegativity decreases down the group.

Reason	Explanation
Atomic radius increases	Outer shell further from nucleus
More shielding	Inner shells block nuclear attraction
Less effective nuclear pull on bonding electrons	Lower electronegativity

Ranking:

F > Cl > Br > I > At

D. Trends in Reactivity (as Oxidising Agents)

Reactivity decreases down the group.

Why:

- Halogens react by **gaining electrons** ($X_2 + 2e^- \rightarrow 2X^-$)
- Down the group:
 - Atomic radius increases
 - More shielding
 - Nuclear attraction to incoming electrons decreases
 - Harder to gain electrons → **lower oxidising power**

Therefore:

- **Cl₂** is the strongest oxidising agent
- **I₂** is the weakest

E. Displacement Reactions (Prove Reactivity Trend)

A more reactive halogen displaces a less reactive halide from solution.

Example:

$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$ (orange solution forms)

Use **cyclohexane** to observe colour change:

- Cl_2 : pale green
- Br_2 : orange
- I_2 : violet

8.25 — Reactivity Trend in Group 7 via Redox Displacement Reactions

A. Core Principle

A **more reactive halogen** will **displace a less reactive halide ion** from its aqueous salt solution.

This confirms the **oxidising ability trend**:

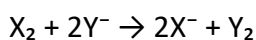
- Chlorine displaces both Br^- and I^-
- Bromine displaces only I^-
- Iodine displaces **neither** Cl^- nor Br^-

B. Explanation in Redox Terms

What's happening:

- The halogen molecule **gains electrons** (is reduced)
- The halide ion **loses electrons** (is oxidised)

General Reaction:



Where X_2 is more reactive than Y_2

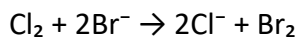
C. Oxidation Number Changes

Species	Oxidation Change
$\text{Cl}_2 \rightarrow 2\text{Cl}^-$	$0 \rightarrow -1$ (reduction)
$2\text{Br}^- \rightarrow \text{Br}_2$	$-1 \rightarrow 0$ (oxidation)

This proves **Cl_2 is acting as the oxidising agent**, and Br^- is the reducing agent.

D. Ionic and Full Redox Equations

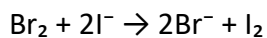
Example 1:



Half equations:

- $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ (reduction)
- $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$ (oxidation)

Example 2:



(No reaction occurs if I_2 is added to Cl^- or Br^-)

E. Colour Changes for Displacement Reactions

Use **cyclohexane layer** to observe colours clearly:

Reaction	Observation in Organic Layer
Cl_2 added to KBr	Orange (Br_2 formed)
Cl_2 added to KI	Violet (I_2 formed)
Br_2 added to KI	Violet (I_2 formed)
I_2 added to KBr/KCl	No change (no reaction)

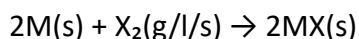
8.26 — Redox Reactions of the Halogens: Oxidation Numbers and Disproportionation

A. Halogen Reactions with Group 1 and 2 Metals

Reaction Type:

Redox — halogen is **reduced**, metal is **oxidised**

General Equation:



Where M = Group 1/2 metal, X = halogen

Example:

- $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$
- $\text{Ca} + \text{Br}_2 \rightarrow \text{CaBr}_2$

Oxidation Numbers:

- Halogen: $0 \rightarrow -1$ (reduction)

- Metal: $0 \rightarrow +1$ or $+2$ (oxidation)

These reactions:

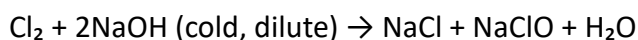
- Form **white crystalline ionic solids**
- Are highly **exothermic**

B. Disproportionation Reactions of Chlorine

Definition:

A disproportionation reaction is one in which the **same element is simultaneously oxidised and reduced**.

1. Chlorine + Cold Dilute NaOH

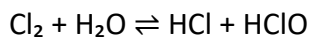


Oxidation numbers:

- Cl_2 :
 - One Cl goes from $0 \rightarrow -1$ in NaCl (reduced)
 - One Cl goes from $0 \rightarrow +1$ in NaClO (oxidised)

This forms **sodium chlorate(I)** (NaClO) $\rightarrow$ the active ingredient in **household bleach**

2. Chlorine + Water

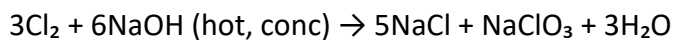


- Cl_2 acts as both oxidising and reducing agent
- HCl: Cl is -1
- HClO: Cl is $+1$

Use:

- Disinfecting drinking water and swimming pools (kills bacteria)

3. Chlorine + Hot Alkali



- Chlorine forms **chlorate(V)** ion (ClO_3^- , Cl = $+5$)

This is another disproportionation reaction, but with a **higher oxidation number** than in bleach (ClO^- , Cl = $+1$)

C. Summary of Disproportionation Reactions

Reaction	Oxidation State Change	Products
$\text{Cl}_2 + \text{H}_2\text{O}$	$0 \rightarrow -1 \text{ and } +1$	$\text{HCl} + \text{HClO}$
$\text{Cl}_2 + \text{cold NaOH}$	$0 \rightarrow -1 \text{ and } +1$	$\text{NaCl} + \text{NaClO}$
$\text{Cl}_2 + \text{hot NaOH}$	$0 \rightarrow -1 \text{ and } +5$	$\text{NaCl} + \text{NaClO}_3$

D. Why Halogens Undergo Disproportionation

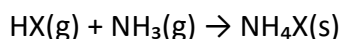
- Halogens are **intermediate in oxidation potential**: they can both gain and lose electrons
- Chlorine in particular can access multiple oxidation states: $-1, 0, +1, +3, +5, +7$

8.27 — Key Reactions of Hydrogen Halides and Their Salts

A. Reaction with Ammonia Gas (NH_3)

Type: Acid–base reaction

General Reaction:



Examples:

- $\text{HCl} + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl}$ (white smoke)
- $\text{HBr} + \text{NH}_3 \rightarrow \text{NH}_4\text{Br}$
- $\text{HI} + \text{NH}_3 \rightarrow \text{NH}_4\text{I}$

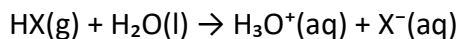
Observation:

- **White smoke or solid** forms
- This is **ammonium halide**, a typical acid-base salt

B. Reaction with Water

Type: Dissociation to form strong acids (except HF, which is weak)

General Reaction:



Strength:

- $\text{HCl}, \text{HBr}, \text{HI} \rightarrow$ **strong acids** (fully dissociate)
- $\text{HF} \rightarrow$ **weak acid** (partially dissociates due to strong H–F bond)

Observation:

- Highly exothermic
- Produces **very acidic** solutions (except HF)

C. Reaction of Solid Group 1 Halides with Concentrated Sulfuric Acid

This is a critical test for identifying halides based on **acid–base** and **redox behaviour**.

1. Chloride (Cl⁻)

- HCl gas released
- No redox occurs
- **Steamy fumes** observed
- **HCl turns damp blue litmus red**

2. Bromide (Br⁻)

Two-step reaction:

- Acid–base:
$$\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HBr(g)}$$
- Redox:
$$2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$$

Observations:

- **Steamy HBr fumes**
- **Brown vapour of Br₂**
- **Pungent SO₂ gas** (choking smell)

Bromide is a **stronger reducing agent** than chloride

3. Iodide (I⁻)

Three reactions:

- Acid–base:
$$\text{NaI} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HI(g)}$$
- Redox (multiple products):
 - $$2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + \text{H}_2\text{O}$$
 - Further reduction: H₂S or S may form

Observations:

- **Steamy HI fumes**
- **Purple vapour of I₂**
- **Smell of rotten eggs (H₂S)**
- **Yellow solid sulfur** may be observed

Iodide is the **strongest reducing agent** among halides

D. Summary: Halide Reducing Ability

Halide Ion	Acid–Base Reaction	Redox Reaction with H ₂ SO ₄	Observations
Cl ⁻	Yes	No	Steamy HCl only
Br ⁻	Yes	Yes	Brown Br ₂ gas, SO ₂ fumes
I ⁻	Yes	Yes (strong)	Purple I ₂ , H ₂ S smell, yellow S

8.28 — Identifying Halide Ions Using Silver Nitrate + Predicting Fluorine and Astatine Reactions**A. Identification of Halide Ions with Silver Nitrate (AgNO₃)**

This test is used to distinguish **Cl⁻, Br⁻, and I⁻** ions in solution.

Procedure:

1. Acidify sample with **dilute nitric acid (HNO₃)**
 - Removes interfering ions like CO₃²⁻ or OH⁻
 - Do **not** use HCl or H₂SO₄ (would add halide ions)
2. Add **silver nitrate solution (AgNO₃)** dropwise
3. Observe precipitate colour
4. Add **aqueous ammonia** (dilute or concentrated) to test solubility

B. Observations and Reactions:

Halide Ion	Precipitate	Colour	Solubility in NH_3 (aq)	Ionic Equation
Cl^-	AgCl(s)	White	Dissolves in dilute NH_3	$\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)}$
Br^-	AgBr(s)	Cream	Dissolves in concentrated NH_3	$\text{Ag}^+ + \text{Br}^- \rightarrow \text{AgBr(s)}$
I^-	AgI(s)	Yellow	Insoluble in NH_3	$\text{Ag}^+ + \text{I}^- \rightarrow \text{AgI(s)}$

Tip: a **white tile** is used for better visibility

C. Predicting Reactions of Fluorine (F_2) and Astatine (At_2)

Fluorine — Above Chlorine in Group 7

Property	Prediction
Oxidising strength	Stronger than Cl_2 (most reactive halogen)
Displacement ability	Can displace Cl^- , Br^- , I^- from solutions
Physical state	Pale yellow gas
HF formation	Reacts violently with water to form HF

Caution:

- Too reactive and toxic for standard school experiments

Astatine — Below Iodine in Group 7

Property	Prediction
Oxidising strength	Weaker than I_2 (least reactive halogen)
Displacement ability	Likely cannot displace I^-
Physical state	Predicted to be solid with metallic appearance
Stability	Radioactive , short-lived isotopes

D. Summary: Halogen Trends

Element	State	Oxidising Power	Colour in Organic Solvent
F_2	Gas	Highest	Very pale yellow (not tested)
Cl_2	Gas	Strong	Pale green
Br_2	Liquid	Moderate	Orange

I ₂	Solid	Weak	Violet
At ₂	Solid (predicted)	Weakest	Dark (unknown)

(Note: Oxidising power decreases down group 7, this means the opposite is also true, the reducing power **increases** down group 7, in exam you are most likely to make either of the two arguments for answering questions)